

Mathematical Strengthening Report for NeurIPS Resubmission

Executive summary

This report focuses on mathematical weaknesses raised during the ICML review cycle and on concrete, rigorous fixes to strengthen theorems, assumptions, proofs, and theoretically-motivated experiments for a NeurIPS resubmission. The analysis is grounded in your current manuscript [filecite]{turn0file0} and the ICML reviews + rebuttals [filecite]{turn0file1}.

The top mathematical weaknesses (and remedies) are:

- 1. Scope mismatch in the “local information leakage” theorem:** The manuscript’s Theorem 3.3 is proved for a *special case* (full-sequence monotone ordering via stars-and-bars) but is used to justify a *general* claim about compliance tasks / event-ordering tasks.
Remedy: Replace Theorem 3.3 with a **general leakage theorem for the actual compliance definition** (Def. 3.1), under explicit assumptions (independence + full support), and demote the stars-and-bars argument to a named special-case proposition (naive setting). Provide a fully rigorous coupling-based proof.
- 2. Overbroad claims (“any ordering-based task...”, “event-level ordering...”) that are false as stated:** A reviewer gave a credible **counterexample family** (induction-head-type constructions) satisfying your “pure sequential dependence” condition while still being order-based.
Remedy: Narrow the claim to the *compliance family you define*, and explicitly acknowledge the existence of non-compliance order-based tasks that can be locally information-free. Add a short theorem/remark exhibiting such a counterexample construction, to prevent future reviewer attacks.
- 3. Definition of “pure sequential dependence” is under-specified mathematically:** The manuscript motivates purity via $I(Y; X_t) = 0$, but “no local shortcuts” is stronger than single-position independence.
Remedy: Upgrade to a **k-local purity definition** (no bounded-window statistic can carry information), and prove that your parity task in the balanced case satisfies purity for *all windows shorter than the full sequence*.
- 4. Parity “unlearnability” interpretation is not theoretically defensible without scaling/controlled ablations:** Reviewers correctly note parity is learnable in simpler regimes; thus failure at ($\ell = 26, |S| = 6, n = 20$) cannot support a universal claim.
Remedy: Reframe as an **empirical failure under standard supervised training** for a *hidden-subset parity* regime; add experiments varying alphabet size ℓ , subset size $|S|$, sequence length n , and providing algorithmic baselines + curricula (with theoretical motivation from parity-learning literature). ¹

5. **“No signal beyond k -grams, no learning” is false as a general statement:** Induction heads are a widely-discussed mechanism enabling beyond- k -gram generalization. ²

Remedy: Restrict the claim to your specific parity variant and training regime; add a short related-work/theory paragraph distinguishing “ k -gram baselines” from algorithmic copying circuits.

Materials used and missing inputs you must provide

The analysis below uses:

- Current manuscript: “Do Large Language Models Detect and Exploit Decisive Sequential Information?”
□filecite□turn0file0□
- ICML reviews and your rebuttals (compiled): □filecite□turn0file1□

To produce *surgical, line-accurate* mapping and propose revised LaTeX-ready proofs, I still need the following missing materials (even if some are “in principle” present, they are not fully recoverable from what I can currently verify):

- The **LaTeX source** (or at least a compiled PDF) of the **supplementary** where you claim the “full proof in supplementary material” for the generalized leakage theorem lives (the rebuttal references an external link). □filecite□turn0file1□
 - The reviewer-provided **parity counterexample code** (the review says it exists; I need the exact setup: model depth/width, optimizer, dataset generation, evaluation). □filecite□turn0file1□
 - Your **dataset generation scripts** for Naive/Tricky/Parity, including the exact parity labeling rule used and whether S is fixed or resampled per example. □filecite□turn0file0□
 - The **exact model configs** used for the parity runs (depth, hidden size, positional encoding choice, tokenization, learning rate schedule, early stopping criterion).
 - The complete, unabridged set of ICML reviews (including any meta-review / AC comments), if the provided PDF is not exhaustive.
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Math-related reviewer comments mapped to manuscript locations

The table below lists each math-related reviewer comment (quote/paraphrase), maps it to the manuscript location, and states what needs to change. When reviewers reference “lines,” the exact line numbers are not consistently recoverable, so I map to section/theorem/equation identifiers.

Reviewer comment (quote/paraphrase)	Type	Manuscript location	Status	What must change (high level)
“Clarify difference between general compliance (Def 3.1) and narrower compliance implicitly used in Theorem 3.3 stars-and-bars proof; current proof doesn’t justify general claim.” □filecite□turn0file1□	Theorem/proof scope	Def. 3.1 (Setup), Thm 3.3 (Pure Sequential Dependence section) □filecite□turn0file0□	Unresolved in manuscript	Replace/rename the theorem; provide a general proof for Def 3.1, move stars-and-bars to special case
“You conflate ‘compliance tasks’ with all event-level ordering tasks; uniform-i.i.d assumption is unrealistic; claims must be qualified.” □filecite□turn0file1□	Assumptions / claim overreach	Abstract + Sec 3.2 conclusion language (“classification tasks based on event-level ordering...”) □filecite□turn0file0□	Partially addressed in rebuttal	Rewrite claims: specify <i>compliance</i> family + distributional assumptions (independence + full support); explicitly do not claim universality
“Induction-head style task can satisfy your ‘pure sequential dependence’ (0 MI per token) yet be order-based, contradicting your broad statement.” □filecite□turn0file1□	Counterexample to broad claim	Sec 3.2 informal implication text + Contributions bullet (“event-based ordering... impossible”) □filecite□turn0file0□	Unresolved in manuscript	Add counterexample remark; narrow theorem scope; distinguish compliance vs associative recall/copying tasks ²
“Parity is learnable on binary vocab; your experiments claim parity unlearnable—needs explanation and scaling evidence.” □filecite□turn0file1□	Empirical-theoretical gap	Def 3.4 + Thm 3.5 + Results “Parity: universal failure” □filecite□turn0file0□	Unresolved in manuscript	Reframe claim; add ℓ -sweep and $ S $ -sweep; include algorithmic baseline; analyze optimization/sample complexity ³

Reviewer comment (quote/paraphrase)	Type	Manuscript location	Status	What must change (high level)
“Claim ‘no signal beyond k -grams’ is incorrect; induction heads show beyond- k generalization.” □filecite□turn0file1□	Claim false as stated	Intro key finding (“No signal beyond k -grams, no learning”) + Discussion tie-in □filecite□turn0file0□	Unresolved in manuscript	Make it task- and training-specific; add related work on induction heads and why your parity differs 4
“Parity supervision is sparse; failure may be optimization/curriculum issue rather than architectural impossibility.” □filecite□turn0file1□	Interpretation / identifiability	Sec 3.3 + Sec 5 parity conclusion language □filecite□turn0file0□	Partially addressed in rebuttal	Replace “fundamental limitation” phrasing with “fails under standard supervised training”; add curriculum/auxiliary-loss experiments 5

Concrete fixes with revised statements, proofs, bounds, and counterexamples

Fix set for Theorem 3.3 scope mismatch

Problem diagnosis

In the manuscript, Theorem 3.3 is stated as an “event-based ordering induces local information” result, and the proof outline counts globally non-decreasing sequences using stars-and-bars. □filecite□turn0file0□ However, Def. 3.1 defines compliance as existence of a lag-spaced chain over a key subset S , which is strictly more general than “the entire sequence is non-decreasing.” □filecite□turn0file0□ Reviewers correctly flag that (a) the proof does not match the general task definition and (b) the manuscript prose overgeneralizes. □filecite□turn0file1□

Concrete manuscript change

Replace the current Theorem 3.3 with a correct general theorem for Def. 3.1 under explicit assumptions, and **rename the current stars-and-bars argument** to a special-case proposition tied to the Naive setting.

Table comparing original vs revised theorem

Item	Current manuscript	Revised proposal
“Theorem 3.3” statement	Ordering-based tasks over finite alphabets induce local information; proof uses globally non-decreasing sequences (stars-and-bars). [filecite turn0file0]	Compliance tasks from Def. 3.1 under independence + full support inevitably leak local information: $\exists a$ s.t. $P(Y^* = 1 \mid X_1 = a)$ differs across a .
Assumptions	i.i.d. uniform, total order κ on full alphabet. [filecite turn0file0]	Independent coordinates; full support $P(X_t = a) > 0$ for all a ; non-degeneracy $0 < \rho < 1$; $n \geq 1 + \lambda$.
Proof technique	Counting monotone sequences. [filecite turn0file0]	Coupling / witness event E with positive probability; show $P(Y^* = 1 \mid X_1 = s_{\min}) > P(Y^* = 1 \mid X_1 = a_0)$. Uses basic MI facts. 6
What happens to stars-and-bars	Used as main theorem proof. [filecite turn0file0]	Becomes “Proposition (Naive full-order case)” supporting intuition and allowing explicit conditional probabilities. 7

Revised theorem and fully rigorous proof

I write this as LaTeX-ready content you can drop into the paper (with minimal editing).

Definition (Compliance; restated). Let $\mathcal{A}$ be a finite alphabet; let $S \subseteq \mathcal{A}$ be the key subset; let $\kappa : S \rightarrow \{1, \dots, |S|\}$ be a bijection defining a strict order on S ; let $\lambda \in \mathbb{N}$, $\lambda \geq 1$. For a sequence $X = (X_1, \dots, X_n)$, define $Y^*(X) = 1$ iff there exist indices $t_1 < \dots < t_k$ with $k \geq 2$ such that (i) $t_{j+1} - t_j = \lambda$, (ii) $X_{t_j} \in S$ for all j , and (iii) $\kappa(X_{t_1}) \leq \dots \leq \kappa(X_{t_k})$. [\[filecite turn0file0\]](#)

Assumption A (Independence + full support). $X_1, \dots, X_n$ are independent random variables over $\mathcal{A}$, and for every $t \in \{1, \dots, n\}$ and every $a \in \mathcal{A}$, $P(X_t = a) > 0$.

Assumption B (Non-degeneracy). The compliance probability $\rho := P(Y^* = 1)$ satisfies $0 < \rho < 1$.

Theorem (Local information leakage for compliance tasks).

Under Def. 3.1 and Assumptions A–B, if $n \geq 1 + \lambda$, then

$$I(Y^*; X_1) > 0.$$

Equivalently, there exist $a, b \in \mathcal{A}$ such that $P(Y^* = 1 \mid X_1 = a) / P(Y^* = 1 \mid X_1 = b)$. [6](#)

Proof. Let $m := |S| \geq 1$. Pick $s_{\min} \in S$ such that $\kappa(s_{\min}) = 1$. We split into two cases.

Case 1: $S \subsetneq \mathcal{A}$. Let $a_0 \in \mathcal{A} \setminus S$ (exists by strict inclusion). Consider the event

$$E := \{X_{1+\lambda} \in S\} \cap \bigcap_{t \in \{2, \dots, n\} \setminus \{1+\lambda\}} \{X_t \in \mathcal{A} \setminus S\}.$$

By Assumption A (full support + independence), $P(E) > 0$, and for any value of X_1 , $P(E \mid X_1 = \cdot) = P(E)$.

- On $E \cap \{X_1 = s_{\min}\}$, the pair of indices $(1, 1 + \lambda)$ is a compliant chain: both symbols lie in S , and $\kappa(X_1) = 1 \leq \kappa(X_{1+\lambda})$. Hence $Y^* = 1$.
- On $E \cap \{X_1 = a_0\}$, exactly one position in $\{1, \dots, n\}$ contains a key symbol (namely $1 + \lambda$); thus there is no chain of length $k \geq 2$ with all symbols in S . Hence $Y^* = 0$.

Therefore,

$$P(Y^* = 1 \mid X_1 = s_{\min}) \geq P(E \mid X_1 = s_{\min}) = P(E) > 0,$$

while

$$P(Y^* = 1 \mid X_1 = a_0) \leq 1 - P(E \mid X_1 = a_0) = 1 - P(E),$$

and in fact, the two conditional probabilities differ because on E the outcomes deterministically differ (1 vs 0). Thus $P(Y^* = 1 \mid X_1 = \cdot)$ is non-constant, implying $I(Y^*; X_1) > 0$ (mutual information is zero iff the variables are independent). ⁶

Case 2: $S = \mathcal{A}$. Then every symbol is “key.” If $X_1 = s_{\min}$, then regardless of the rest of the sequence, the length-2 chain $(1, 1 + \lambda)$ is compliant because $\kappa(X_1) = 1 \leq \kappa(X_{1+\lambda})$. Hence

$$P(Y^* = 1 \mid X_1 = s_{\min}) = 1.$$

By Assumption B, $\rho = P(Y^* = 1) < 1$. If $P(Y^* = 1 \mid X_1 = a) = 1$ for all $a \in \mathcal{A}$, then by total probability $P(Y^* = 1) = 1$, contradicting $\rho < 1$. Hence there exists $a^* \in \mathcal{A}$ with $P(Y^* = 1 \mid X_1 = a^*) < 1$. Again $P(Y^* = 1 \mid X_1 = \cdot)$ is non-constant, so $I(Y^*; X_1) > 0$. ■

What this fixes. This theorem matches Def. 3.1 exactly (including subset S and lag λ), and it removes the incorrect dependence on “globally non-decreasing sequences” as the main proof.

Optional strengthening: an explicit lower bound (useful in exposition)

If you want more than “ > 0 ,” under i.i.d. uniform you can lower bound $P(E)$ explicitly:

$$P(E) = \left(\frac{m}{\ell}\right) \left(\frac{\ell - m}{\ell}\right)^{n-2},$$

so

$$P(Y^* = 1 \mid X_1 = s_{\min}) - P(Y^* = 1 \mid X_1 = a_0) \geq P(E).$$

This gives a *quantitative* “leakage magnitude” you can plot versus (m, ℓ, n) , tightening the story beyond existence.

Fix set for claim overreach and the induction-head counterexample

Problem diagnosis

The manuscript makes statements paraphrasable as:

- “Any ordering-based classification task over a finite alphabet necessarily leaks information...”
□filecite□turn0file0□
- “Classification tasks based on event-level ordering cannot satisfy pure sequential dependence...”
□filecite□turn0file0□

A reviewer disputes this, and the concern is correct in principle: there exist order-dependent tasks whose label is independent of every individual token, including induction-head-style constructions associated with in-context learning. □filecite□turn0file1□ 4

Your rebuttal attempts to resolve this by narrowing “ordering-based” to “compliance tasks (Def 3.1),” which is the right direction but must be made airtight in the manuscript. □filecite□turn0file1□

Concrete manuscript changes

1. Replace **global-language claims** with scoped claims:
2. Replace “any ordering-based task” → “any *compliance task* as in Def. 3.1 under independence + full support...”
3. Add a short **Remark / Proposition (Counterexample family)**:
4. “There exist order-based tasks with $I(Y; X_t) = 0$ for each t , so our impossibility result is specific to compliance tasks, not all order-based tasks.”

A clean counterexample you can include

This is a mathematically crisp version (not tied to transformers) that demonstrates the logical point the reviewer is making.

Proposition (Order-based task with zero per-token mutual information).

Let $\mathcal{A} = \{0, 1, \dots, \ell - 1\}$ and let $n \geq 4$. Draw $X_1, \dots, X_n$ i.i.d. uniform on $\mathcal{A}$. Define the label

$$Y := (X_2 + X_3) \bmod \ell.$$

Then for every $t \in \{1, \dots, n\}$, $I(Y; X_t) = 0$, yet Y depends on the sequence order (it is a function of positions 2 and 3, not a bag-of-tokens statistic).

Proof sketch. Since X_2, X_3 are i.i.d. uniform, their sum mod ℓ is uniform; for any fixed x_2 , $(x_2 + X_3) \bmod \ell$ is uniform, so $Y \perp X_2$, and similarly $Y \perp X_3$. For $t \notin \{2, 3\}$, Y is a measurable function of (X_2, X_3) only, so $Y \perp X_t$ by independence. Hence $I(Y; X_t) = 0$ for all t . ■

How to use this in your paper. Put it right after the scoped leakage theorem to clarify: your impossibility is about *compliance tasks*, not about “any order-based label.”

Connection to induction heads

Induction heads are often modeled as a “match-and-copy” mechanism $[A][B] \dots [A] \mapsto [B]$. ⁸

This supports the reviewer’s claim that transformers can realize beyond- k -gram mechanisms, and it motivates why “no signal beyond k -grams” cannot be globally true. ²

Fix set for “pure sequential dependence” definition

Problem diagnosis

Your Def. 3.2 informally rules out frequency/co-occurrence/order-invariant properties, and you motivate it with $I(Y; X_t) = 0$. ^{filecite{turn0file0}}

But if the goal is to formalize “no local shortcut,” independence from each single X_t is often too weak: a 2-gram or small window could still leak information even if each marginal is independent.

Concrete fix: define k -local purity

Definition (k -local purity). A binary classification task (X, Y) over sequences $X \in \mathcal{A}^n$ is *k -locally pure* (w.r.t. distribution P) if for every window $W_t = (X_t, \dots, X_{t+k-1})$ with $t \leq n - k + 1$,

$$I(Y; W_t) = 0.$$

Equivalently, no function of any contiguous length- k window can predict Y better than chance.

This captures your “ k -gram baseline has no signal” idea as a property of the data distribution rather than an artifact of a particular baseline.

Strengthen the parity theorem accordingly

Your manuscript’s Theorem 3.5 proves $I(Y^*; X_t) = 0$ in the balanced case $p = |S|/|\mathcal{A}| = 1/2$. ^{filecite{turn0file0}}

You can strengthen this substantially:

Theorem (Parity is k -locally pure for all $k < n$ when $p = 1/2$).

Let $B_1, \dots, B_n$ be i.i.d. Bernoulli(1/2) and let $T = (\sum_{t=1}^n B_t) \bmod 2$. Then for any subset of indices $J \subset \{1, \dots, n\}$ with $|J| < n$, we have $I(T; (B_j)_{j \in J}) = 0$. In particular, for any contiguous window length $k < n$, $I(T; B_{t:t+k-1}) = 0$.

Proof. Let J^c be the complement; since $|J^c| \geq 1$, the parity of $\sum_{t \in J^c} B_t$ is uniform on $\{0, 1\}$ and independent of $(B_j)_{j \in J}$. But $T = (\sum_{t \in J} B_t + \sum_{t \in J^c} B_t) \bmod 2$, i.e., the XOR of a known bit (given $(B_j)_{j \in J}$) with a uniform independent bit, hence uniform independent of the conditioning. Therefore $T \perp (B_j)_{j \in J}$, giving zero mutual information. ■

This aligns your mathematical notion of “no local shortcut” with what reviewers intuitively mean by “parity requires global computation.”

Fix set for parity: reframing claims + adding mathematically motivated experiments

Problem diagnosis

The manuscript uses parity failure to suggest a “fundamental limitation” (or at least a strong general barrier) about global integration. [filecite\[turn0file0\]](#)

Reviewers correctly point out parity is learnable in simpler settings (binary vocab), and that failure may reflect optimization/sample complexity rather than representational impossibility. [filecite\[turn0file1\]](#)

The deeper mathematical issue: *learning parity-like functions with gradient-based methods can be fragile and regime-dependent*, and there is a substantial learning-theory literature about parity hardness in various models. [3](#)

Concrete reframing edits (drop-in text)

Replace any phrasing like “parity is unlearnable for transformers” with:

“In the hidden-subset parity regime we study ($\ell = 26, |S| = 6, n = 20$), none of the tested architectures learned above chance **under standard supervised training** with cross-entropy and conventional optimization. This should be interpreted as an empirical limitation of the training setup and inductive biases in this regime, not as a proof of impossibility.”

This matches reviewer concerns while keeping your result valuable. [filecite\[turn0file1\]](#)

Add an algorithmic baseline (to separate learnability from optimization)

Add a non-neural baseline that solves parity exactly:

- Learn a mapping $f : \mathcal{A} \rightarrow \{0, 1\}$ where $f(a) = 1$ iff $a \in S$ (if S is fixed globally).
- Predict $\hat{Y} = \left(\sum_{t=1}^n f(X_t) \right) \bmod 2$.

This baseline has **AUC = 1** (no noise) and demonstrates solvability. It also lets you quantify how much of the difficulty is “subset identification” vs “parity computation.”

Add a controlled scaling suite (must-have for NeurIPS)

To directly answer the strongest reviewer critiques, add the following experiments (each is theoretically motivated and can be summarized in 1–2 figures):

1. **Alphabet size sweep:** $\ell \in \{2, 4, 8, 16, 26, 52\}$ with matched $p = |S|/\ell = 1/2$.
Hypothesis: learning sharpens for small ℓ and degrades with larger ℓ if optimization is the bottleneck (consistent with parity-learning difficulty depending on representation and gradient signal). [9](#)
2. **Subset size sweep:** keep $\ell = 26$, vary $|S| \in \{1, 2, 3, 6, 13\}$ (including balanced $|S| = 13$).
Point: identify whether the issue is “hidden subset identification” (harder when $|S|$ near $\ell/2$) or “parity” itself.

3. **Sequence length sweep:** $n \in \{10, 20, 40, 80\}$.

Motivation: parity is known to stress long-range dependencies; length scaling can differentiate representational vs optimization limitations (and connects to classic long-term dependency issues).

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4. **Curriculum and auxiliary losses** (directly responsive to reviewer zMcq):

5. Curriculum on n : train on small n then increase.

6. Auxiliary prediction of the running parity $(\sum_{i \leq t} f(X_i)) \bmod 2$ at intermediate layers or at multiple positions (deep supervision).

Curriculum learning for parity has explicit mathematical treatments showing regime sensitivity. 5

7. **Known-subset vs hidden-subset parity ablation:**

8. Known-subset: provide $f(X_t)$ as an input feature (bit) to the model; tests pure parity computation.

9. Hidden-subset: raw letters only; tests subset identification + parity.

This directly reconciles “binary parity is easy” with “26-letter hidden-subset parity is hard.”

Add theory-aware discussion referencing parity-learning literature

A short paragraph in Related Work / Discussion should mention:

- Learning parity functions is a canonical learning-theory stress test; several works analyze why gradient-based training struggles in some parity regimes and how curricula/initializations change this. 11
- “Parity with noise” (LPN) is computationally hard in general and foundational in cryptography; while you do **not** study LPN, it reinforces that parity-like problems can be regime-sensitive. 12

Fix set for “no signal beyond k-grams” claim

Problem diagnosis

The manuscript’s headline claim “No signal beyond k-grams, no learning” is stated as if it were general across transformers/LLMs. filecite turn0file0

Reviewers correctly flag this is false in general, given induction heads and in-context learning behavior. filecite turn0file1 8

Concrete fix

Replace the general claim with a precise empirical statement:

“In our hidden-subset parity task at $\ell = 26, n = 20$, fixed- k k-gram statistics carry no detectable signal, and under standard supervised training, none of the tested models exceeded chance. This reflects the compositional difficulty of this task regime rather than a general limitation of transformers.”

Also add a one-paragraph related work pointer:

- Induction heads are a mechanism enabling beyond- k -gram generalization in transformers. ¹³
- Therefore, “ k -gram failure implies transformer failure” is not logically valid; your result is about a particular distribution/task family.

Proposed proof and exposition upgrades beyond reviewer demands

Even beyond the direct complaints, two mathematical upgrades will make the paper more NeurIPS-robust.

Upgrade: make distributional assumptions explicit everywhere they matter

Currently, the manuscript intertwines: (i) benchmark generation (uniform i.i.d letters), (ii) theory about leakage, (iii) noisy labels. `filecite{turn0file0}`

Make a “Assumptions box” early in Section 3:

- **A0 (Benchmark distribution):** i.i.d uniform on $\mathcal{A}$ (only for data generation + closed-form ρ).
- **A1 (Leakage theorem distribution):** independence + full support (weaker than uniform and better matched to reviewer concerns).
- **A2 (Noise model):** symmetric label noise with $\pi < 1/2$, with a citation to standard classification noise literature. ¹⁴

This eliminates reviewer claims that you are “extrapolating from uniform” when your core theorem can be stated more generally.

Upgrade: sharpen Theorem 4.1 exposition (“optimal AUC under label noise”)

Your theorem is correct and already fairly complete. `filecite{turn0file0}`

Two improvements:

1. Explicitly state the **Bayes-optimal scoring function** given access to Y^* is $s(X) = \mathbb{P}(Y = 1 \mid Y^*)$, which is a two-point score (one value for $Y^* = 1$, one for $Y^* = 0$). This makes the AUC derivation conceptually cleaner.
2. Cite the classification noise model as standard (Angluin–Laird). ¹⁵

Experiment suite to validate theoretical claims and strengthen the NeurIPS narrative

This section translates the proof fixes into experiments that reviewers typically demand when theory touches empirical claims.

Experiments tied to Theorem “compliance leaks local information”

1. **Non-uniform i.i.d distributions:** Replace uniform letters with a Zipf-like marginal over $\mathcal{A}$, keeping compliance definition unchanged. (This tests reviewer concerns that uniformity is unrealistic.) The generalized leakage theorem under full support predicts leakage persists under *any* full-support marginals; empirical k-gram signal should remain nonzero. ⁶
2. **Independent but non-identical positions:** Let position-dependent marginals P_t vary; leakage theorem still holds under independence. Show leakage magnitude changes but remains >0 .
3. **Algorithmic baseline vs k-gram:** For compliance tasks, include a baseline that explicitly checks compliance given (S, κ, λ) (oracle) and show its AUC matches Theorem 4.1 bound under noise; this triangulates correctness. ⁵

Experiments tied to parity learning dispute

Add a figure with 3 panels:

- Panel A: Learnability heatmap over $(\ell, |S|)$ at fixed n .
- Panel B: Effect of known-subset vs hidden-subset.
- Panel C: Curriculum / auxiliary losses (show at least one method that improves or show convincingly that multiple fail, to support the claim “hard under standard training”).

The relevant conceptual backdrop: gradient-based methods can struggle on parity-like targets in some regimes; curricula can help in others. ⁵

Mermaid dependency diagrams and table of proposed manuscript changes

```
flowchart TB
    A[Def 3.1: Compliance task] --> B[Assumptions: independence + full support]
    B --> C[Thm: Compliance leaks local info  $I(Y^*;X_1)>0$ ]
    C --> D[Cor: k-gram / local baselines can have signal]
    A --> E[Label noise model  $\pi$ ]
    E --> F[Thm 4.1: Bayes-optimal AUC/F1 given  $Y^*$ ]
    G[Def: k-local purity] --> H[Parity task (balanced  $p=1/2$ )]
    H --> I[Thm: k-local purity for all  $k<n$ ]
    I --> J[Empirical test: hidden-subset parity]
    J --> K[Interpretation: failure under standard training (not impossibility)]
    L[Counterexample: order-based but locally  $MI=0$ ] --> M[Scope clarification]
    M --> C
```

Table summarizing proposed manuscript changes

Manuscript part	Proposed change	Purpose	Reviewer issue addressed
Abstract + Contributions	Replace “any ordering-based task...” with “compliance tasks (Def 3.1) under independence + full support...” □filecite□turn0file0□	Prevent incorrect universality claim	Overreach critique □filecite□turn0file1□
Section 3.2 (Leakage theorem)	Replace Thm 3.3 with the coupling-based general theorem; demote stars-and-bars to “Naive-case proposition” □filecite□turn0file0□	Align statement, proof, and use in narrative	Thm 3.3 mismatch □filecite□turn0file1□
Add remark/proposition	Include explicit counterexample of order-based task with $I(Y; X_t) = 0$	Immunize against induction-head counterexample and clarify scope	Induction-head critique □filecite□turn0file1□
Definition 3.2	Add k-local purity definition (optional but recommended)	Formalize “no local shortcuts” beyond token marginals	Clarity/rigor issue
Section 3.3 + parity discussion	Strengthen theorem to “k-local purity for all $k < n$ ” in balanced case; reframe empirical claim as training-regime failure	Improves theoretical clarity and avoids overclaiming	Parity + “fundamental limitation” critique □filecite□turn0file1□
Experiments (new)	Add ℓ -sweep, $ S $ -sweep, n -sweep, known-vs-hidden subset, curriculum/auxiliary loss	Empirically reconcile binary parity learnability with your failure regime	Parity learnability dispute □filecite□turn0file1□
Intro / Results claim	Replace “no signal beyond k-grams, no learning” with task-specific statement; add induction-head related work	Fix false general claim	k-gram critique □filecite□turn0file1□ 16

Core primary sources to cite in the revised NeurIPS manuscript

To strengthen the scholarly grounding for the math-facing claims and reviewer discussions, the following are the highest-leverage citations:

- Induction heads as a mechanism for in-context learning: 8
- Beyond-transformer sequence modeling context referenced by reviewers (induction heads / ICL ecosystem): 17
- Classification noise model foundations: 15
- Noise-tolerant learning + parity relevance: 18
- Parity learning and gradient-based difficulties / regimes: 19
- Information theory fact “MI=0 iff independence” for your arguments: 6

This completes the mathematically focused strengthening plan and provides LaTeX-ready revised theorems/proofs, scope corrections, counterexamples, and an experiment suite that directly addresses the ICML reviewer critiques while strengthening the NeurIPS submission.

- 1 3 19 **Learning Parities with Neural Networks**
https://proceedings.neurips.cc/paper/2020/file/eaee5e04a259d09af85c108fe4d7dd0c-Paper.pdf?utm_source=chatgpt.com
- 2 4 8 13 16 **In-context Learning and Induction Heads**
https://arxiv.org/abs/2209.11895?utm_source=chatgpt.com
- 5 **A Mathematical Model for Curriculum Learning for Parities**
https://proceedings.mlr.press/v202/cornacchia23a/cornacchia23a.pdf?utm_source=chatgpt.com
- 6 **Elements of Information Theory**
https://cs-114.org/wp-content/uploads/2015/01/Elements_of_Information_Theory_Elements.pdf?utm_source=chatgpt.com
- 7 **Stars and bars (combinatorics)**
https://en.wikipedia.org/wiki/Stars_and_bars_%28combinatorics%29?utm_source=chatgpt.com
- 9 11 **Hardness of Learning Fixed Parities with Neural Networks**
https://arxiv.org/abs/2501.00817?utm_source=chatgpt.com
- 10 **Learning long-term dependencies with gradient descent is ...**
https://www.comp.hkbu.edu.hk/~markus/teaching/comp7650/tnn-94-gradient.pdf?utm_source=chatgpt.com
- 12 **Cryptographic Primitives Based on Hard Learning Problems**
https://kilthub.cmu.edu/articles/Cryptographic_Primitives_Based_on_Hard_Learning_Problems/6604574/files/12094979.pdf?utm_source=chatgpt.com
- 14 15 **Learning From Noisy Examples**
https://homepages.math.uic.edu/~lreyzin/papers/angluin88b.pdf?utm_source=chatgpt.com
- 17 **Mamba: Linear-Time Sequence Modeling with Selective State Spaces**
https://arxiv.org/abs/2312.00752?utm_source=chatgpt.com
- 18 **Noise-Tolerant Learning, the Parity Problem, and the ...**
https://www.microsoft.com/en-us/research/wp-content/uploads/2016/11/2003-Noise-Tolerant_Learning.pdf?utm_source=chatgpt.com